

An Efficient Scene Coordinate Encoding and Relocalization Method

Kuan Xu^{1*}, Zeyu Jiang^{1*}, Haozhi Cao¹, Shenghai Yuan¹, Chen Wang², Lihua Xie¹, *Fellow, IEEE*

Abstract—Scene Coordinate Regression (SCR) is a visual localization technique that utilizes deep neural networks (DNN) to directly regress 2D-3D correspondences for camera pose estimation. However, current SCR methods often face challenges in handling repetitive textures and meaningless areas due to their reliance on implicit triangulation. In this paper, we propose an efficient scene coordinate encoding and relocalization method. Compared with the existing SCR methods, we design a unified architecture for both scene encoding and salient keypoint detection, enabling our system to focus on encoding informative regions, thereby significantly enhancing efficiency. Additionally, we introduce a mechanism that leverages sequential information during both map encoding and relocalization, which strengthens implicit triangulation, particularly in repetitive texture environments. Comprehensive experiments conducted across indoor and outdoor datasets demonstrate that the proposed system outperforms other state-of-the-art (SOTA) SCR methods. Our single-frame relocalization mode improves the recall rate of our baseline by 6.4% and increases the running speed from 56Hz to 90Hz. Furthermore, our sequence-based mode increases the recall rate by 11% while maintaining the original efficiency.

I. INTRODUCTION

Visual relocalization is crucial for many applications such as mobile robots and augmented reality, offering a practical balance between cost and accuracy [1]. Its core task involves estimating the camera pose using an existing map, which can be broadly divided into two categories: explicit maps and implicit maps. Explicit maps consist of a 3D point cloud combined with visual descriptors of 2D features [2], [3], while implicit maps encode the scene using a neural network [1], [4]. Feature matching (FM)-based methods construct explicit maps using techniques like structure-from-motion (SfM) [3], [5] or simultaneous localization and mapping (SLAM) [2], [6]. During relocalization, these methods rely on image retrieval and feature matching to establish 2D-3D correspondences between pixels in the query image and 3D points in the map. However, this approach often suffers from large map sizes due to the storage of visual features and 3D models. Additionally, the process can be computationally inefficient, as the feature matching needs to be performed between the query image and multiple candidates.

*Equal contribution.

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¹Kuan Xu, Zeyu Jiang, Haozhi Cao, Shenghai Yuan, and Lihua Xie are with the Centre for Advanced Robotics Technology Innovation (CARTIN), School of Electrical and Electronic Engineering, Nanyang Technological University, 50 Nanyang Avenue, Singapore 639798, {kuan.xu, zjiang015, haozhi002, shyuan, elhxie}@ntu.edu.sg.

²Chen Wang is with Spatial AI & Robotics Lab, Department of Computer Science and Engineering, University at Buffalo, Buffalo, NY 14260 chenw@sairlab.org.

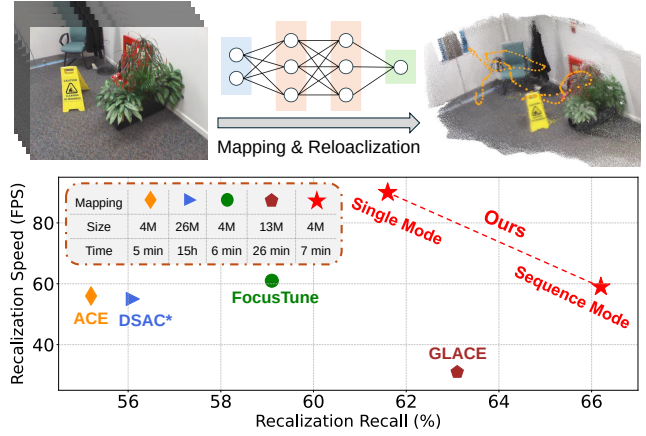


Fig. 1: In this paper, we propose an efficient SCR system that leverages deep neural networks for both mapping and relocalization. Our system delivers accurate and efficient relocalization, while maintaining fast mapping and a compact map size, achieving state-of-the-art performance.

In contrast, scene coordinate regression (SCR) methods encode map information into a neural network [1], [7]. They directly regress the 3D coordinates in the world coordinate system correspondences to the image pixels, eliminating the need for explicit 2D-3D corresponding searching. Compared to FM-based methods, SCR methods enable faster relocalization with a smaller map. Recently, ACE [8] proposes a generalized convolutional neural network (CNN) to extract feature maps for scene encoding. Building on this pre-trained backbone, it utilizes a compact 4MB-sized multilayer perceptron (MLP) head as the map, enabling new scene encoding in five minutes only without requiring a 3D model or depth information. This approach achieves state-of-the-art accuracy and efficiency.

Despite their advantages, SCR methods face several notable limitations. First, the presence of many texture-less and non-informative regions in the image significantly increases the map encoding errors [9]. Second, existing RGB-only SCR methods rely heavily on implicit triangulation, i.e., the neural network must consistently identify and map the same regions across different images to accurate 3D points. This reliance causes a sharp decline in performance in environments with repetitive textures [10]. Third, during relocalization, all pixel-to-3D matches are considered for pose estimation since the reliability of specific regions is unknown, which significantly reduces efficiency. Efforts have been made to address these challenges. FocusTune [9] mitigates encoding errors by focusing on informative regions but requires a 3D model of the scene, which limits its

practicality. For the second issue, GLACE [10] introduces global image descriptors to better distinguish similar regions across images, yet this approach comes at the cost of reduced mapping and localization efficiency.

In this paper, we present a novel SCR system designed to tackle these challenges. To address the first and third issues, we introduce a unified and scene-agnostic CNN backbone that simultaneously encodes the scene and identifies salient regions, focusing mapping and relocalization efforts exclusively on these informative areas. For the second challenge, we explicitly associate the salient regions across different images, leveraging sequential information to enhance implicit triangulation and relocalization. In summary, the main contributions of this work are as follows:

- We introduce a unified, scene-agnostic CNN backbone for scene encoding and salient region detection, which mitigates accuracy degradation caused by non-informative areas.
- We propose a novel method that leverages sequential information to strengthen the implicit triangulation in SCR systems, improving mapping accuracy, particularly in environments with repetitive textures.
- We design two relocalization modes: a single-frame mode that delivers one-shot relocalization with high efficiency and a sequence mode that utilizes temporal information to enhance relocalization performance.
- Extensive experimental results show that our methods achieve substantial improvement in both efficiency and effectiveness. We provide the source code at <https://github.com/sair-lab/SeqACE>.

II. RELATED WORKS

A. Feature Matching Based Relocalization

FM-based visual relocalization methods remain the state-of-the-art for large-scale and challenging environments [10]. These approaches rely on 3D models generated through SfM or SLAM [2], [3], [11], [12]. During relocalization, candidate map images similar to the query image are retrieved using techniques such as bag-of-words (BoW) or image retrieval methods [3], [6]. Feature matching is then performed between the query image and each retrieved map image. Since the keypoints in the map images have already been triangulated, 2D-3D correspondences between the pixels in the query image and the 3D points in the point cloud model can be established. Finally, the 6-DOF pose of the query image is estimated using geometric techniques, such as Perspective-n-Point (PnP) and bundle adjustment [13], [14].

However, FM-based methods require storing 2D keypoints, visual descriptors, and a 3D point cloud, resulting in relatively large map sizes. For large-scale scenes, managing these maps demands significant memory resources. Furthermore, during relocalization, multiple candidate frames are typically retrieved for feature matching, which imposes computational overhead and reduces efficiency.

B. Absolute Pose Regression

Absolute pose regression (APR) approaches use a DNN to regress camera poses from RGB images directly. They have also garnered notable attention recently due to their good efficiency and simple architecture. PoseNet [15] is the first ARP solution which uses GoogLeNet [16] as the backbone. The following works explore different architectural designs to enhance ARP systems [17], [18]. Other works leverage sequential information to improve APR accuracy by incorporating temporal filtering [19] or visual odometry [20]. Recent ARP works utilize novel synthesis techniques to generate high-quality training data [21] or integrate them for fine-tuning [22]. However, ARP approaches still fall short in accuracy compared to SCR and FM-based methods.

C. Scene Coordinate Regression

Different from ARP approaches, SCR methods regress the corresponding 3D coordinates in the scene for 2D pixels in the query image. Most existing SCR methods require depth information or a 3D model of the scene. Dong *et al.* [7] utilize a pre-learned feature extractor and scene region classifier to encode a scene with few-shot images, achieving efficient mapping. However, the additional modules, e.g., feature detection and hierarchical classification, limit its localization speed to only 5Hz even on a powerful GPU. Zhou *et al.* [23] propose KFNet. They leverage the sequential information by integrating the traditional Kalman filter [24] into the SCR pipeline. Bui *et al.* [25] propose D2S, which builds on a pre-trained sparse feature extraction network. They establish a mapping between feature descriptors and 3D scene points.

DSAC++ [26] is the first SCR method that does not rely on depth or 3D model information. However, it requires a long training time. DSAC* [1] simplifies the DSAC++ training procedure by unifying two separate initialization steps, reducing the training time from 6 days to 15 hours. ACE [8] replaces the backbone of DSAC* with a pre-trained CNN, requiring only a new header to be trained when encoding a new scene, which greatly improves the mapping efficiency. This reduces the mapping time to 4 minutes and the map size to 4MB. FocusTune [9] and GLACE [10] are variations of ACE. FocusTune projects a 3D model onto training images to avoid encoding meaningless image regions, but this introduces the dependency on the 3D model. GLACE incorporates global image descriptors into the ACE pipeline to enhance the distinguishability of similar regions.

III. PRELIMINARIES

A. Problem Statement

The objective is to estimate the pose $\mathbf{T}_{wc}$ of a given image I , where $\mathbf{T}_{wc}$ denotes the transformation from the camera coordinate system to the world coordinate system. Given a set of 2D-3D correspondences $\mathcal{C} = (\mathbf{p}_i, \mathbf{P}_i^w)$, where $\mathbf{p}_i$ represents a pixel in the image I and $\mathbf{P}_i^w$ is its corresponding 3D point in the world coordinate system, the pose can be efficiently computed using PnP algorithms or iterative optimization techniques. To obtain these correspondences $\mathcal{C}$,

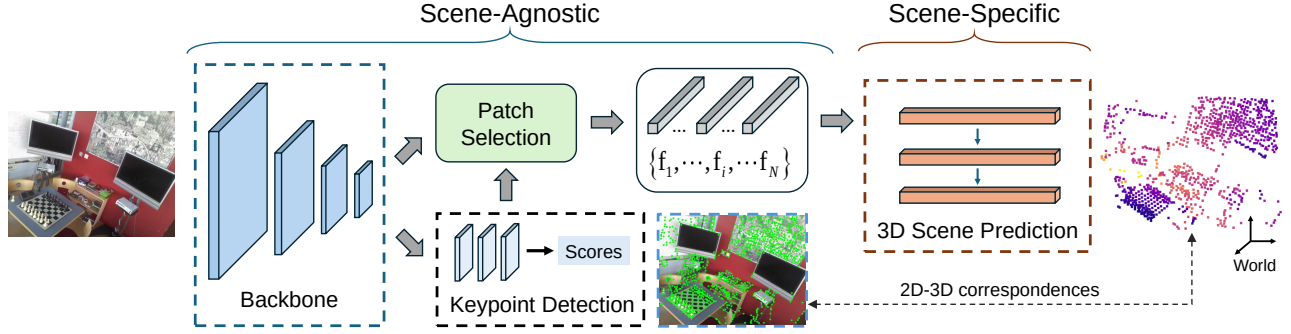


Fig. 2: Our system comprises a scene-agnostic module and a scene-specific module. The scene-agnostic module identifies informative image patches and generates high-dimensional features for each patch. The scene-specific module, implemented using 1D convolutional layers, predicts the corresponding 3D scene point for each patch. These 2D-3D correspondences are then utilized for relocalization.

SCR methods aim to learn a mapping from 2D pixels to their corresponding 3D points:

$$\mathbf{P}_i^w = f(\mathcal{P}(\mathbf{p}_i, I); \mathbf{w}), \quad (1)$$

where $\mathcal{P}(\mathbf{p}_i, I) \in \mathbb{R}^{C_p \times H_p \times W_p}$ is a small patch centered at $\mathbf{p}_i$ in the image I , and f is a neural network with learnable parameters $\mathbf{w}$. Existing SCR pipelines usually take grayscale inputs with $C_p = 1$. As there is no explicit patch extraction, the patch size is determined by the receptive field of the neural network. Considering the trade-off between invariance and discriminative power, the patch size is typically set to $H_p = W_p = 81\text{px}$ in many existing architectures [8], [10].

B. Accelerated Coordinate Encoding

Training f in Eq. 1 using only posed images takes a long time [1]. To accelerate this process, Accelerated Coordinate Encoding (ACE) [8] proposes to split f into a scene-agnostic backbone f_B and a scene-specific MLP head f_H :

$$\mathbf{P}_i^w = f_H(\mathbf{f}_i; \mathbf{w}_H), \text{ with } \mathbf{f}_i = f_B(\mathcal{P}(\mathbf{p}_i, I); \mathbf{w}_B), \quad (2)$$

where f_B predicts a high-dimensional feature vector $\mathbf{f}_i$ for each patch, and f_H maps $\mathbf{f}_i$ to the corresponding 3D point:

$$f_B: \mathbb{R}^{C_p \times H_p \times W_p} \rightarrow \mathbb{R}^{C_f}, \text{ and } f_H: \mathbb{R}^{C_f} \rightarrow \mathbb{R}^3, \quad (3)$$

where C_f is the dimension of $\mathbf{f}_i$. The backbone f_B is pre-trained on a large dataset. When mapping a new scene, only the MLP head f_H needs to be trained while f_B remains fixed. Thus, the MLP head serves as the map. This approach significantly reduces both the map size and the mapping time.

IV. METHODS

A. Overview

The overall structure of the proposed system is depicted in Fig. 2. It comprises two main components: a scene-agnostic module and a scene-specific module. The scene-agnostic module generates feature vectors and saliency scores for image patches, while the scene-specific module, designed as an MLP head with several 1D convolutional layers, maps these feature vectors to 3D scene points. The detailed design of the scene-agnostic module is presented in Section IV-B. Section IV-C describes how sequential information is utilized

to enhance the training of the scene-specific module, and Section IV-D outlines our relocalization methods.

B. Scene-Agnostic Module

In this section, we aim to design an efficient and scene-agnostic module capable of (1) detecting informative image patches and (2) mapping these patches into high-dimensional features for scene encoding. As many learning-based patch and keypoint detection methods have been proposed [3], [27], a straightforward solution is to utilize them directly for salient region detection along with the ACE backbone. However, employing two separate deep neural networks would significantly reduce online efficiency. Inspired by [3], [6], we think that these two tasks can be accomplished using a shared backbone. To achieve this, we extend the ACE backbone by adding a lightweight keypoint detection head (KDH). The KDH consists of just three convolutional layers, introducing trivial computational overhead. Its input is the feature map from the backbone, and its output is a tensor $\mathcal{T}_K \in \mathbb{R}^{65 \times \frac{H}{8} \times \frac{W}{8}}$, where H and W denote the height and width of the input image, respectively. The 65 channels correspond to an 8×8 grid region and a dustbin channel representing the absence of keypoints. This tensor is processed through a softmax layer and resized into a $H \times W$ heatmap, which scores the saliency of the corresponding image pixels. Since the KDH, like the backbone f_B in Eq. 2, is also scene-agnostic, we combine them into a unified module denoted as f_S , with its parameters represented by $\mathbf{w}_S$. Then the second equation in Eq. 2 is updated to:

$$\mathbf{f}_i, \mathcal{S}_i = f_S(\mathcal{P}(\mathbf{p}_i, I); \mathbf{w}_S), \quad (4)$$

where $\mathcal{S}_i$ is the salient score of $\mathbf{p}_i$.

We train f_S using a two-step pipeline on the ScanNet dataset [28]. In the first step, we follow the approach outlined in [8] to train the backbone f_B . In the second step, the parameters of f_B are fixed, and we utilize distillation techniques to train the KDH. Theoretically, any keypoint detector could serve as the teacher network. For our setup, we use the prediction of SuperPoint [27] as supervision signals. The loss function is defined as:

$$\mathcal{L}_K = \mathcal{L}_{CE}(\mathcal{T}_K, \mathcal{T}'_K), \quad (5)$$

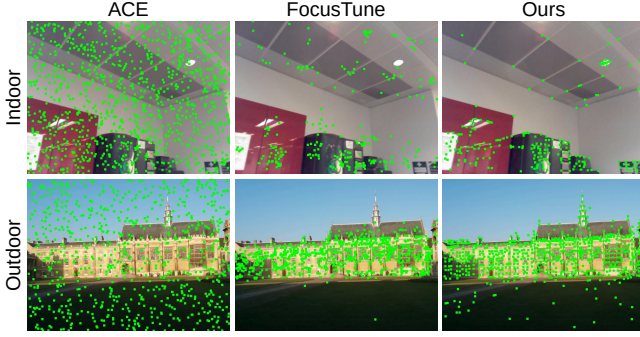


Fig. 3: Sampling strategies comparison. ACE randomly samples image patches. FocusTune performs uniform sampling around the projected points of the 3D model. We use keypoint detection to select repeatable image patches.

where $\mathcal{L}_{CE}$ is the cross-entropy loss and $\mathcal{T}'_K$ is the prediction of the teacher network. We also experimented with a multi-task learning approach [3] to train f_S , but found that the two-step pipeline produced better results. This is probably because the difficulty of the two tasks differs significantly: specifically, training a keypoint detector is much simpler than training a universal scene encoder. This disparity makes it challenging to balance the two tasks effectively.

C. Sequence-Based Mapping

Saliency-Based Buffer Sampling Buffer sampling is the key to fast mapping. Since the backbone, i.e., f_B or f_S , is scene-agnostic, it only needs to be executed once when mapping a new scene. The resulting outputs, $\{\mathbf{f}_i\}$, can be cached and subsequently used to train f_H . To enhance efficiency, not all image patches are utilized during training. In ACE [8], 1024 image patches are randomly sampled per image, and their corresponding $\mathbf{f}_i$, $\mathbf{p}_i$, camera intrinsics, and camera poses are stored in a training buffer. However, this strategy may sample many meaningless regions. To address this, FocusTune [9] projects the 3D point cloud of the scene onto the training images and performs uniform sampling around the projected points. While this approach better targets informative areas, it cannot guarantee consistent sampling across multiple images.

In contrast, we utilize the keypoint detection results from Section IV-B to select the top 1,000 patches with the highest confidence scores in each image, aligning with the number used by ACE. This approach ensures consistent patch selection across image sequences. It also ensures the same patches can be used during mapping and relocalization. We show a comparison of patch selection in Fig. 3.

Self-Loss We follow the previous works [8], [10] to compute a reprojection loss to supervise the training of f_H :

$$\mathcal{L}_{S,i} = \begin{cases} \tau(t) \tanh\left(\frac{e_\pi(\mathbf{p}_i, \mathbf{P}_i^w, \mathbf{T}_{wc})}{\tau(t)}\right) & \text{if } \mathbf{P}_i^w \in \mathcal{V} \\ \|\mathbf{P}_i^w - \bar{\mathbf{P}}_i^w\| & \text{otherwise,} \end{cases} \quad (6)$$

where $\mathbf{P}_i^w$ is the 3D scene point predicted by f_H using $\mathbf{f}_i$ and $e_\pi(\mathbf{p}_i, \mathbf{P}_i^w, \mathbf{T}_{wc})$ is the reprojection error. $\mathcal{V}$ is a set of valid predictions that are between 0.1m and 1000m in front of the image plane and have a reprojection error of less than

1000px. $\bar{\mathbf{P}}_i^w$ is the pseudo ground truth 3D point computed by the inverse projection of $\mathbf{p}_i$ with the camera pose $\mathbf{T}_{wc}$ and a fixed target depth at 10m. $\tau(t)$ is a dynamically adjusted scale between $\tau_{\min} = 1$ and $\tau_{\max} = 50$:

$$\tau(t) = \tau_{\max} \sqrt{1 - t^2} + \tau_{\min}, \quad (7)$$

where $t \in (0, 1)$ represents the relative training progress.

For each image patch, $\mathcal{L}_{S,i}$ uses information from a single image to supervise the prediction of $\mathbf{P}_i^w$ for each $\mathbf{f}_i$, so we call it self-loss. However, in classical geometry, a single viewpoint constrains a scene point to lie along a line of sight rather than determining its precise 3D position. For instance, $\mathcal{L}_{S,i}$ would be zero if the predicted $\mathbf{P}_i^w$ falls anywhere on the inverse-projection line of $\mathbf{p}_i$. Therefore, SCR systems heavily rely on effective implicit triangulation, i.e., the neural network must consistently identify the same region across multiple images so that the scene point can be constrained by observations from different viewpoints. Achieving this is particularly challenging in complex or repetitive texture environments, as shown in [10].

Cross-Loss To tackle the above challenge, we propose explicitly integrating multi-view constraints into the mapping process, rather than relying solely on the neural network to learn them implicitly. The core idea is to associate the selected patches across different images using feature-matching techniques [29]. This approach enables us to project a predicted 3D point onto multiple frames and leverage the resulting reprojection errors for more effective supervision.

Inspired by visual SLAM techniques, we select keyframes to ensure that matching patch pairs have varying baselines. A frame is chosen as a keyframe if the matched features with the previous keyframe are less than half of the detected features. For each matching pair $(\mathbf{p}_{k,j}, \mathbf{p}_i)$, where $\mathbf{p}_{k,j}$ is the matched point on the keyframe, we define a loss as:

$$\mathcal{L}_{C,i} = \begin{cases} \tau(t) \tanh\left(\frac{e_\pi(\mathbf{p}_{k,j}, \mathbf{P}_i^w, \mathbf{T}_{wk})}{\tau(t)}\right) & \text{if } \mathbf{P}_i^w \in \mathcal{V} \\ \|\mathbf{P}_i^w - \bar{\mathbf{P}}_{k,j}^w\| & \text{otherwise,} \end{cases} \quad (8)$$

where $\mathbf{T}_{wk}$ is the pose of the keyframe and $\bar{\mathbf{P}}_{k,j}^w$ is the pseudo ground truth 3D point computed by the inverse projection of $\mathbf{p}_{k,j}$. Since $\mathcal{L}_{C,i}$ uses different frames to supervise the prediction of scene points, we call it cross-loss.

Training Details For each image patch, we store the corresponding feature vector $\mathbf{f}_i$, the pixel $\mathbf{p}_i$, the camera intrinsic $\mathbf{K}$, the camera pose $\mathbf{T}_{wc}$, the pixel matched on the keyframe $\mathbf{p}_{k,j}$, and the keyframe pose $\mathbf{T}_{wk}$ in the training buffer. We shuffle these patches and select $\mathcal{N}_b = 5120$ patches in each iteration to compute the loss:

$$\mathcal{L} = \frac{1}{\mathcal{N}_b} \sum (\mathcal{L}_{S,i} + \lambda \mathcal{L}_{C,i}), \quad (9)$$

where $\lambda = 0.5$ if $(\mathbf{p}_{k,j}, \mathbf{p}_i)$ is an inlier match, otherwise $\lambda = 0$. We train f_H over 16 complete iterations using the AdamW optimizer [30] with a learning rate between $5e-4$ and $5e-3$ and a 1-cycle schedule [31].

TABLE I: Comparison on the 7-Scenes dataset. The relocalization recall is the proportion of test frames below a (1cm, 1°) pose error.

		Mapping Time	Map Size	FPS*	Relocalization Recall (%)							
					Chess	Fire	Heads	Office	Pumpkin	Kitchen	Stairs	Avg.
FM	GoMatch [13]	1.5h	~300M	1.23	1.6	1.4	0	0.6	0.3	0.5	0.7	0.7
	PixLoc [14]	1.5h	~1.7G	1.23	89.1	66.2	77.6	55.3	56.7	66.5	9.8	60.2
	Hloc (SP+SG) [3]	1.5h	~3.5G	0.46	96.1	71.2	82.5	68.6	75.2	79	16.3	69.8
SCR	DSAC* [1]	15h	26.3M	55	82.3	59.3	85.4	39.4	56.4	67.6	2.6	56.1
	ACE [8]	5 min	4.1M	56	87.2	60.1	87.9	46.2	34.3	66.7	3.8	55.2
	FocusTune [9]	6 min	4.1M	61	90.4	64.3	90.1	49.6	47.1	68.3	3.9	59.1
	GLACE [10]	18 min	12.8M	31	87.4	67.1	86.3	60.5	58.7	75.8	5.8	63.1
	Ours-Single	7 min	4.1M	90	94.2	63.5	87.9	56.2	51.9	67.7	11.0	61.6
	Ours-Sequence	7 min	4.1M	59	96.5	68.8	91.7	59.8	57.8	71.7	17.4	66.2

* Running time of relocalization measured in frame per second (FPS).

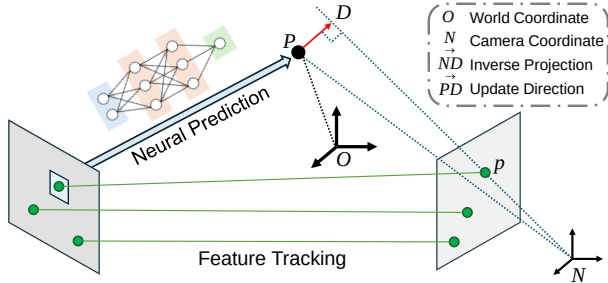


Fig. 4: In our sequence-based relocalization, the predicted 3D scene points are used to constrain the pose estimation of subsequent frames and updated with new observations.

D. Dual-Mode Relocalization

In this section, we introduce how to reuse the map for relocalization. Our system supports two relocalization modes. The first is single-mode, which estimates the camera pose using a single image. This mode is efficient and straightforward. As shown in Fig. 2, the framework generates a set of 2D keypoints and their corresponding predicted 3D scene points. Using these 2D-3D correspondences, we apply PnP and RANSAC algorithms to compute the optimal pose by maximizing the number of inliers. Like most existing SCR methods, our single-mode does not leverage the constraints between adjacent frames in an image sequence. However, in real-world robotic applications, the previous frame provides valuable prior information for pose estimation. Therefore, we also design a sequence-mode that utilizes sequential information to enhance the relocalization. The details of this mode are introduced in the following section.

Pose Prediction In traditional visual odometry methods [32], local constraints are established by first triangulating 3D points from historical frames and then using these points to constrain the current pose. Typically, triangulation involves solving an $\mathbf{Ax} = \mathbf{b}$ problem through matrix decomposition, which can be computationally intensive. To simplify this process, we propose a more efficient approach. Our key insight is that if the reprojection errors of a predicted scene point across multiple historical frames remain small, the prediction can be considered accurate and used to constrain the current pose. Based on this idea, we maintain a set of reliably predicted scene points for pose estimation.

We denote two adjacent frames as $\mathcal{I}_m$ and $\mathcal{I}_n$, where $\mathcal{I}_n$

is the current frame and $\mathcal{I}_m$ is the previous frame. Let $\mathcal{Y}_m$ and $\mathcal{X}_m$ represent the maintained set of 3D scene points and their corresponding 2D points on $\mathcal{I}_m$, respectively. Using optical flow [33], we track $\mathcal{X}_m$ to obtain $\bar{\mathcal{X}}_n$ on $\mathcal{I}_n$, along with their corresponding scene points $\bar{\mathcal{Y}}_n$ (denoted with a bar to indicate pre-update values). The pose of $\mathcal{I}_n$ is then computed as:

$$\bar{\mathbf{T}}_{wn} = \mathbf{g}(\bar{\mathcal{X}}_n, \bar{\mathcal{Y}}_n), \quad (10)$$

where $\mathbf{g}$ represent the PnP solver with RANSAC.

Pose Update The pose estimation in Eq. 10 only uses 2D-3D correspondences of historical frames and feature tracking results, so $\bar{\mathbf{T}}_{wn}$ can be refined with new predictions from the network for $\mathcal{I}_n$. Let $\hat{\mathcal{X}}_n$ and $\hat{\mathcal{Z}}_n$ denote the newly detected keypoints and the corresponding 3D scene points predicted by the network. Since the set of 2D-3D correspondences $\{\hat{\mathcal{X}}_n, \hat{\mathcal{Z}}_n\}$ typically contains many outliers, we first use PnP with RANSAC to compute a pose $\hat{\mathbf{T}}_{wn} = \mathbf{g}(\hat{\mathcal{X}}_n, \hat{\mathcal{Z}}_n)$, then fuse it with $\bar{\mathbf{T}}_{wn}$ at the pose level, rather than directly using a Kalman filter. We perform a simple weighted average on the manifold to obtain the updated pose:

$$\mathbf{T}_{wn} = \exp \left(\left(\bar{w}_n \log(\bar{\mathbf{T}}_{wn})^\vee + \hat{w}_n \log(\hat{\mathbf{T}}_{wn})^\vee \right)^\wedge \right), \quad (11)$$

where $\exp$ and $\log$ represent the exponential and logarithmic maps, respectively. The weights $\bar{w}_n$ and $\hat{w}_n$ are computed as:

$$\bar{w}_n = \frac{\bar{\mathcal{N}}_n}{\bar{\mathcal{N}}_n + \hat{\mathcal{N}}_n}, \quad \hat{w}_n = \frac{\hat{\mathcal{N}}_n}{\bar{\mathcal{N}}_n + \hat{\mathcal{N}}_n} \quad (12)$$

where $\bar{\mathcal{N}}_n$ and $\hat{\mathcal{N}}_n$ are the number of inliers when estimating $\bar{\mathbf{T}}_{wn}$ and $\hat{\mathbf{T}}_{wn}$ using $\mathbf{g}$.

Scene Point Update The maintained scene point set also requires updating based on new observations from $\mathcal{I}_n$. Besides adding newly detected points and their corresponding predicted 3D points to $\{\bar{\mathcal{X}}_n, \bar{\mathcal{Y}}_n\}$, we also update tracked scene points. For a scene point $\bar{\mathbf{P}}_i^w \in \bar{\mathcal{Y}}_n$ and its corresponding tracked 2D point $\bar{\mathbf{p}}_i^n \in \bar{\mathcal{X}}_n$, if the pair $(\bar{\mathbf{p}}_i^n, \bar{\mathbf{P}}_i^w)$ is an outlier in Eq. 10, it will be removed from $\bar{\mathcal{Y}}_n$ and $\bar{\mathcal{X}}_n$. Otherwise, $\bar{\mathbf{P}}_i^w$ is updated in a direction that reduces the reprojection error on $\mathcal{I}_n$. The update direction is illustrated in Fig. 4, where P and p represent $\bar{\mathbf{P}}_i^w$ and $\bar{\mathbf{p}}_i^n$ for simplicity. The inverse projection line of $\bar{\mathbf{p}}_i^n$ is denoted as $\overrightarrow{ND}$, with D

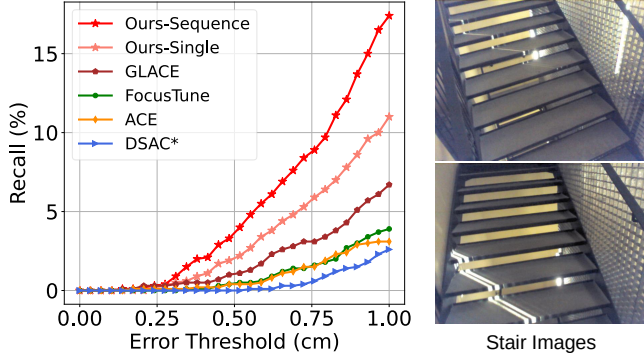


Fig. 5: The recall rates in repetitive texture environments as the threshold varies from 0 to 1cm (left) and the scene images (right).

as the foot of the perpendicular. The coordinates of D are:

$$\bar{\mathbf{D}}_i^w = \left((\bar{\mathbf{P}}_i^w - \mathbf{t}_{wn})^\top \mathbf{v}_{nd} \right) \mathbf{v}_{nd} - (\bar{\mathbf{P}}_i^w - \mathbf{t}_{wn}), \quad (13a)$$

$$\mathbf{v}_{nd} = \mathbf{R}_{wn} \frac{\mathbf{K}_n^{-1} \bar{\mathbf{p}}_i^n}{\|\mathbf{K}_n^{-1} \bar{\mathbf{p}}_i^n\|}, \quad (13b)$$

where $\mathbf{R}_{wn}$ and $\mathbf{t}_{wn}$ are the rotation and translation components of $\mathbf{T}_{wn}$, respectively. $\mathbf{v}_{nd}$ is the direction vector of $\overrightarrow{ND}$. Then $\bar{\mathbf{P}}_i^w$ can be updated through:

$$\mathbf{P}_i^w = \bar{\mathbf{P}}_i^w + \beta_n (\bar{\mathbf{D}}_i^w - \bar{\mathbf{P}}_i^w), \quad (14)$$

where β_n is a scale factor. As $\mathbf{P}_i^w$ is observed across more frames, the influence of each new observation should diminish, allowing $\mathbf{P}_i^w$ to stabilize gradually. Hence, we set $\beta_n = 1/N_P$, where N_P denotes the number of observations.

V. EXPERIMENTAL RESULTS

In this section, we present the experimental results. The experiments are performed on three standard datasets for SCR tasks: 7-Scenes [34], 12-Scenes [35], and Cambridge Landmarks [15]. These datasets provide training data for mapping and test data for relocalization. We evaluate the two relocalization modes introduced in Section IV-D. In the following sections, “Ours-Single” refers to the single-mode relocalization, while “Ours-Sequence” or “Ours” refers to the sequence-based relocalization. All experiments are conducted on an NVIDIA GeForce RTX 3090 GPU.

A. Comparison with State-of-the-Art

7-Scenes The 7-Scenes dataset comprises RGB-D images captured in seven indoor scenes using a Kinect RGB-D camera with a resolution of 640×480 . We train f_H using the RGB frames and their ground-truth poses. Our approach is evaluated against several SOTA SCR systems and FM-based methods. Key metrics, including mapping time, map size, relocalization speed, and recall rate, are evaluated. The recall rate is defined as the percentage of test frames with pose errors below (1cm, 1°). Results are summarized in Table I. Our single-mode relocalization achieves remarkable efficiency, running at 90Hz, which is $1.6\times$ faster than ACE, while also improving its recall by 6.4%. The sequence mode further delivers SOTA accuracy with high efficiency. By comparison, FocusTune and GLACE, both variants of ACE,

TABLE II: Comparison on the 12-Scenes dataset. We report the percentage of test frames below a (1cm, 1°) pose error.

Sequence	DSAC*	ACE	FocusTune	GLACE	Ours
Apt1_kitchen	87.1	84.9	88.0	93.3	97.5
Apt1_living	91.3	89.5	85.2	92.9	95.9
Apt2_bed	89.3	83.6	88.5	88.5	90.2
Apt2_kitchen	83.0	84.3	90.0	93.0	90.9
Apt2_living	83.6	89.1	89.4	91.1	91.6
Apt2_luke	66.5	76.4	77.1	82.7	84.8
Office1_gates362	73.1	73.6	76.2	94.6	81.1
Office1_gates381	62.3	68.2	70.0	85.2	78.8
Office1_lounge	75.8	70.6	69.1	82.3	81.7
Office1_manolis	70.9	72.9	71.4	84.4	77.2
Office2_5a	62.6	60.2	54.1	68.6	73.8
Office2_5b	58.3	71.4	61.2	84.0	85.9
Average	75.3	77.1	76.7	86.7	85.7
FPS	48	53	56	33	60

TABLE III: Comparison on the Cambridge Landmarks dataset. Metrics include the median pose error and the recall rate.

Sequence	Median Error (cm / $^\circ$)				Recall (%)			
	ACE	Ours	$\downarrow \uparrow$		ACE	Ours	$\downarrow \uparrow$	
GreatCourt	43 / 0.2	38 / 0.2	-5 / 0		6.7	9.5	+2.8	
KingsCollege	28 / 0.4	25 / 0.4	-3 / -0		11.7	12.8	+1.1	
OldHospital	31 / 0.6	27 / 0.6	-4 / 0		10.4	9.9	-0.5	
ShopFacade	5 / 0.3	5 / 0.3	-0 / -0		85.4	86.4	+1.0	
StMarysChurch	18 / 0.6	17 / 0.6	-1 / 0		20.2	23.0	+2.8	
Average	25 / 0.4	22 / 0.4	-3 / -0		26.9	28.3	+1.4	

have notable drawbacks: FocusTune relies on a 3D model and achieves only a modest accuracy gain (+3.9%), while GLACE sacrifices mapping and relocalization efficiency. In contrast, our system achieves a substantial recall improvement (+11%), while preserving ACE’s fast mapping time, compact map size, and relocalization speed.

The proposed method significantly enhances system performance in environments with repetitive textures. The “Stairs” sequence is captured on a staircase with numerous repetitive patterns, which poses a significant challenge for accurate relocalization. In this scenario, our single-mode and sequence-mode approaches improve the recall rate of ACE by 7.2% and 13.6%, respectively. This highlights the effectiveness of the sequence-based mapping and relocalization methods, contributing recall rate improvements of 8.9% and 4.8%, respectively. Fig. 5 illustrates scene images and the cumulative error distribution from 0 to 1cm for different systems under this challenging scenario.

12-Scenes The 12-Scenes dataset, collected in 12 scenes using an iPad color camera, provides RGB sequences at a resolution of 1296×968 . Like previous methods, we resize the image to 642×480 during relocalization. We evaluate our system alongside several SOTA SCR approaches, with results summarized in Table II. Compared to the baseline (ACE), our methods improve relocalization recall by 8.6% and boost relocalization speed by 7Hz. Although GLACE achieves slightly higher accuracy on this dataset, it comes at a significant cost: requiring $2.5\times$ the mapping time, consuming $3\times$ the memory for map storage, and operating at only half the relocalization speed of our approach. In Fig. 6, we compare the trajectory errors and recall rates of different

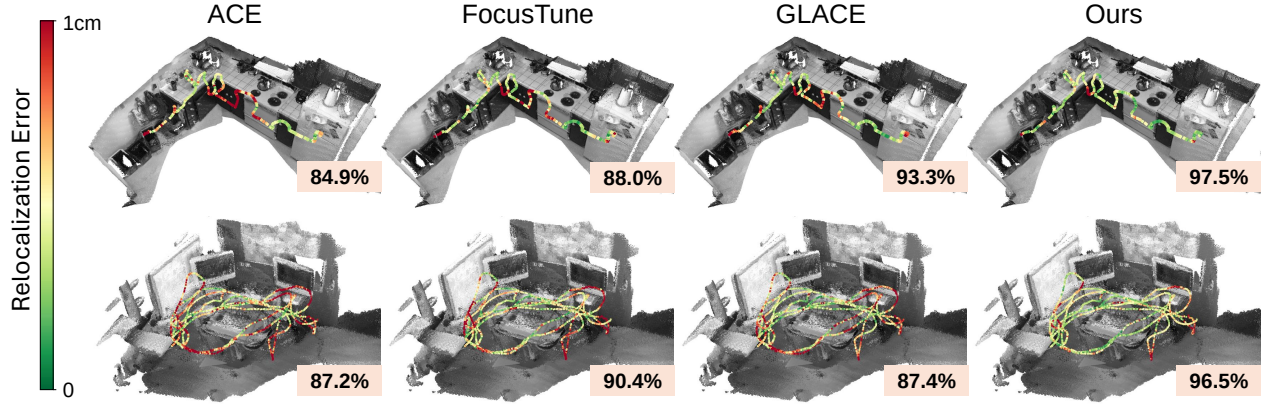


Fig. 6: The relocalization trajectories of different SCR systems in two scenarios (top: Apt1_kitchen, bottom: Chess). We also report the recall rates, defined as the percentage of test frames below a (1cm, 1°) pose error.

TABLE IV: Comparison with other sequential SCR methods. We report the median pose error (cm / °) and the running speed.

Sequence	Cai <i>et al.</i> [36]	Tang <i>et al.</i> [37]	KFNet [23]	Ours
Chess	2.0 / 0.8	2.0 / 0.7	1.8 / 0.7	0.5 / 0.1
Fire	2.0 / 1.0	2.0 / 0.8	2.3 / 0.9	0.7 / 0.3
Heads	4.0 / 2.7	1.0 / 0.8	1.4 / 0.8	0.4 / 0.2
Office	3.0 / 0.8	3.0 / 0.8	2.5 / 0.7	0.9 / 0.2
Pumpkin	4.0 / 1.1	4.0 / 1.1	3.7 / 1.0	0.9 / 0.2
Kitchen	4.0 / 1.1	3.0 / 1.1	3.8 / 1.2	0.7 / 0.2
Stairs	18.0 / 3.9	4.0 / 1.6	3.3 / 0.9	2.4 / 0.7
Average	5.3 / 1.6	4.0 / 1.2	2.7 / 0.9	0.9 / 0.3
FPS	-	15	20	59

algorithms for relocalization in two scenes.

Cambridge Landmarks This dataset comprises five scenes from the old town of Cambridge, captured using a Google LG Nexus 5 smartphone. The captured videos were down-sampled into low-frame-rate image sequences, leading to large motion gaps between consecutive frames. These gaps make feature tracking via optical flow challenging, so we evaluate only our single-mode relocalization on this dataset. The results, summarized in Table III, compare our system with ACE to highlight the benefits of incorporating sequential information during mapping. Metrics include the median pose error and the recall, defined as the percentage of test frames below a (10cm, 10°) pose error. Our methods reduce pose error by 3cm and improve recall by 1.4%, demonstrating the effectiveness of leveraging sequential information.

B. Comparison with Sequential Methods

We further compare our method with other SCR approaches that leverage sequential information, using the 7-Scenes dataset for evaluation. The median pose errors and running speeds of each system are summarized in Table IV. Our system outperforms all others, achieving the best accuracy across all seven scenes with 67% lower translation error and 67% lower rotation error compared to the second-best method. This improvement likely stems from the fact that the other methods encode all pixels, including non-informative or misleading regions, which can dilute the network’s encoding capacity and hinder mapping accuracy. Additionally, [36] introduces an extra module for dense feature extraction,

TABLE V: Ablation study on the proposed keypoint detection head, sequence-based mapping, and sequence-based relocalization.

	7-Scenes Recall (%)		12-Scenes Recall (%)	
w/o KDH, SM, SR	55.2		77.1	
w/o SM, SR	60.3	+5.1	80.4	+3.3
w/o SR	61.6	+6.4	81.4	+4.3
Ours (Full)	66.2	+11.0	85.7	+8.6

while [37] and KFNet [23] rely on dense matching, which compromises efficiency. In contrast, our method employs a lightweight head for detecting sparse features and efficiently tracks them using optical flow, striking a better balance between accuracy and computational cost.

C. Ablation Study

We conduct an ablation study on the 7-Scenes and 12-Scenes datasets to evaluate the contributions of the proposed keypoint detection head (KDH), sequence-based mapping (SM), and sequence-based relocalization (SR). By sequentially removing these three modules, we calculate the recall rate, defined as the percentage of test frames below a (1cm, 1°) pose error. The results, presented in Table V, clearly demonstrate the effectiveness of the proposed components. Since our system builds upon ACE, the configuration “w/o KDH, SM, SR” corresponds to the original ACE. On the 7-Scenes dataset, KDH, SM, and SR improve the recall rate by 5.1%, 1.3%, and 4.6%, respectively, achieving a total improvement of 11.0%. Similarly, on the 12-Scenes dataset, these methods lead to an 8.6% increase in recall rate, further validating their impact.

VI. CONCLUSION

In this work, we propose an efficient and accurate SCR system. Our approach introduces a unified scene-agnostic module for both scene encoding and salient keypoint detection, enabling the system to prioritize encoding informative regions. To strengthen implicit triangulation, we associate keypoints across multiple images to effectively leverage sequential information. We further design two relocalization

modes: the single-mode, which performs one-shot relocalization with high efficiency, and the sequence-mode, which incorporates temporal information to boost relocalization performance. Experimental results demonstrate that our methods significantly improve system performance, achieving remarkable accuracy and efficiency. Nevertheless, like other SCR systems, we use a CNN as the backbone, which limits scene point predictions to rely solely on information from a single patch. In future work, we plan to explore other architectures like ViT [38] to enhance SCR systems.

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